



Samsung PNE8000 plasma series

into the background, becoming less visible to the user and morphing with the environment.

"In the last few years, we have seen apps like email, Twitter and Facebook on consumer electronics devices like TVs," DuBravac said. "Manufacturers wanted to show these properties because it said to consumers that the product is connected. But the experience is generally poor. The next focus will be on improving the user experience."

DuBravac also noted that companies will replace complexity with simplicity, and also that natural interfaces like gesture and voice control will show up in more and more devices such as tablets during 2012.

User interface is a key area of innovation today as it can add value to an existing device, meet the special needs of a new device and make life easier for users. Here let us look at some recent innovations in this space.

Wave a wand or chant a spell

Last year was the year of touch with devices sporting touchscreens and multi-touch capabilities ruling the roost. Computer operating systems such as Windows 8 and Mac OS X Lion too feature a touchscreen-inspired interface as against the traditional desktop environment. This shows that touch is gradually becoming the main interface for not just mobile devices but primary computing devices too.

Now device makers seem to be go-

ing one step ahead, towards friction-less and more natural user interfaces such as gesture and voice.

Microsoft Kinect, Apple's iPad (since October 2011) and iPhone (with Siri) are some brilliant examples. A lot of smartphone makers are including voice-enabled search and other such features in their phones. More recently, Samsung and other consumer electronics companies have also entered the smart interaction game.

Samsung's new PNE8000 plasma series, and the UNES7500 and UNES8000 LED-based LCDs TVs have a built-in camera and microphone, so you can control the television by speaking or gesturing to it. Two remotes are also included. One is a standard multi-button type, while the other is a touchpad with a few buttons and a microphone.

But, voice training is a painful exercise with most voice-enabled devices like these. The system takes too long to train to the user's voice. Repeated instructions to raise the volume might end up opening Skype—found a reviewer!

Gesture control appears to be a bit better. In the Samsung model, it is activated by holding a hand up in front of the TV. Then, you need to move your hand to guide a cursor, grab with your fist to make a selection, or make

an anti-clockwise circular motion to undo. However, you need to have perfect, shadow-free lighting for gesture control.

LG has been showing off its magic remote controls for quite a few years now, every year adding more features to it. Last year, the company demonstrated gesture control for televisions. LG's remote works similar to the Nintendo Wii, where users can use certain hand and arm gestures to operate the onscreen menus. For example, if you gesture a 'V' in the air, it brings up a list of recently watched videos. A wave of the remote can also cause the TV to switch from 2D to 3D and vice versa.

This year, the company has added voice recognition for text input, a scroll wheel and more magic gestures. The voice control system allows you to search for TV and Web content, and also access social media content. The voice features are powered by a platform from Nuance—the maker of Dragon Dictate, a popular Mac tool.

Both Samsung and LG have made a decent attempt at gesture and voice control but a lot more fine-tuning is needed before this goes mainstream.

Speaking of remotes, Canonical has launched a remote-less television called Ubuntu TV. Dubbed the television for human beings, Ubuntu TV has no remote. Users can work with touch or gesture control, or use their smartphone as a remote. The TV runs on the Ubuntu Linux operating system. It includes most Internet TV features like DVR abilities, access to online movies and TV shows, streaming capabilities and intelligent search. Ubuntu

One, Ubuntu's own cloud, will also be built into the new TV so that users can watch movies and TV shows stored on other devices.

Last year, a group of four students in Ahmedabad developed an innovative device called the Air Glove. The



Remote-less Ubuntu TV by Canonical